

# AI Driven Intelligent Rubber Production Monitoring System

1<sup>st</sup> Bhanuka K.Y.

*Faculty of Computing*

*Sri Lanka Institute of Information  
Technology*

Malabe, Sri Lanka

IT22563750@my.sliit.lk

2<sup>nd</sup> Hewamanna K.T.

*Faculty of Computing*

*Sri Lanka Institute of Information  
Technology*

Malabe, Sri Lanka

IT22584090@my.sliit.lk

3<sup>rd</sup> Wadanambi P.W.

*Faculty of Computing*

*Sri Lanka Institute of Information  
Technology*

Malabe, Sri Lanka

IT22355928@my.sliit.lk

4<sup>th</sup> Appuhamy M.H.M.M.

*Faculty of Computing*

*Sri Lanka Institute of Information  
Technology*

Malabe, Sri Lanka

IT22625298@my.sliit.lk

5<sup>th</sup> Mr. Ravi Supunya

*Faculty of Computing*

*Sri Lanka Institute of Information  
Technology*

Malabe, Sri Lanka

ravi.s@sliit.lk

6<sup>th</sup> Mrs. Dulani Jayasinghe

*Faculty of Computing*

*Sri Lanka Institute of Information  
Technology*

Malabe, Sri Lanka

dulani.j@sliit.lk

**Abstract**—The rubber industry in Sri Lanka, which supports more than 200,000 smallholder families, is always having problems. These include losing crops to diseases and pests, having to guess at prices, and not being able to trace the supply chain. This paper presents the design, implementation, and evaluation of an AI-driven intelligent rubber production monitoring system that unifies six integrated modules within a single mobile platform: (i) deep-learning-based disease, pest, and weed detection using ONNX-optimised convolutional neural network models with GPS-tagged geospatial outbreak alerting; (ii) automated rubber sheet quality grading into RSS standards via image classification; (iii) machine-learning-based rubber price forecasting with a blockchain-backed real-time auction marketplace; (iv) a Digital Product Passport (DPP) employing AES-256-CBC zero-knowledge encryption and QR code anti-fraud verification for end-to-end supply chain traceability; (v) IoT-based environmental stress monitoring using ESP32 sensors for soil moisture, humidity, and temperature; and (vi) an NLP-powered agricultural chatbot for rubber cultivation advisory. The system uses React Native (Expo) for cross-platform mobile delivery, ASP.NET Core 8.0 with MongoDB for the backend API, and ONNX Runtime for inference on the device and on the server. Field testing with 40 rubber industry stakeholders in three districts showed that the system had a 91.4% accuracy rate in classifying diseases, an 89.7% accuracy rate in grading, an MAE price prediction of LKR18.3 per kilogram, and a System Usability Scale score of 79.2/100. The system provides a scalable and technically sound solution for modernising rubber production management in Sri Lanka, supporting the advancement of precision agriculture practices.

**Index Terms**—rubber production, deep learning, ONNX, disease detection, rubber grading, price forecasting, blockchain, digital product passport, IoT, ESP32, supply chain traceability, zero-knowledge encryption, mobile application, precision agriculture, Sri Lanka.

## I. INTRODUCTION

Rubber (*Hevea brasiliensis*) is a highly significant plantation crop in Sri Lanka, sustaining the livelihoods of more than 200,000 smallholder farming families and making a considerable contribution to national export revenues [1]. Notwithstanding its significance, the industry encounters systemic inefficiencies at each phase of the production value chain, encompassing plantation health monitoring, quality assessment, market pricing, and product traceability.

At the plantation level, biotic hazards such as leaf diseases (white root disease, anthracnose, powdery mildew, and leaf spot), pests (scale insects, mites, and caterpillars), and weed competition result in yield losses estimated at 30%–40%. Traditional identification depends on costly professional consulting, which is predominantly unattainable for isolated rural farmers [3]. Post-harvest, the quality grading of rubber sheets into Ribbed Smoked Sheet (RSS) categories is a manual and subjective procedure prone to inter-grader variability.

The opacity of price negotiation and the lack of standardised traceability methods diminish the competitiveness of Sri Lankan rubber in quality-sensitive foreign markets [4].

Advancements in deep learning, edge computing, IoT, and blockchain technology offer an opportunity to address these challenges through an intelligent monitoring platform. Lightweight CNN architectures designed for mobile inference now facilitate real-time AI classification on standard hardware, in conjunction with the Open Neural Network Exchange (ONNX) runtime [8]. Blockchain-based product passports and zero-knowledge encryption can create unalterable trust mechanisms for supply chain integrity, ensuring that all transactions are verifiable and that product information remains secure and tamper-proof throughout the supply chain process.

This paper demonstrates the design, implementation, and evaluation of an AI-driven intelligent rubber production monitoring system that integrates six modules into a centralised mobile platform:

- 1) This paper's remaining sections are organised as follows: Related work is reviewed in Section II. Section III explains methodology and system architecture. Section IV: specifics of execution. Results are shown in Section V and the assessment. Section VI discusses findings and limitations. Section VII concludes the paper.

## II. RELATED WORK

### A. AI-Based Plant Disease Detection

Deep learning has made significant progress in identifying plant diseases. This is especially true since Mohanty et al. [1] showed that convolutional neural networks (CNNs) could achieve over 99% accuracy on the PlantVillage dataset. Withana [2] then created a CNN-based system for detecting rubber tree diseases in Sri Lanka. Although this system achieved 87% accuracy, it was limited to a specific set of disease categories. Wang *et al.* [7] proposed GMA-Net for rubber leaf disease recognition at 92% accuracy, emphasising region-specific training data requirements.

### B. Agricultural Quality Assessment

Automated crop quality assessment using computer vision has been explored for rubber latex characteristics [12], rice [11], and tea [10]. Conversely, image-based grading of rubber sheets into RSS categories has received comparatively little attention, and existing systems predominantly depend on subjective, labour-intensive manual sensory evaluation, thereby hindering precise quality determination.

### C. Price Prediction and Blockchain in Agriculture

Machine learning regression models for agricultural commodity pricing have been utilised in the rubber [13] and palm oil markets. Blockchain-based supply chain traceability has been investigated for food [9] and agricultural products; however, its implementation in the rubber industry is still in its early stages. Kim *et al.* [14] highlighted that NFT-based product passports can be used to check the authenticity of a supply chain.

### D. Confidentiality-Aware Digital Product Passport

Digital Product Passports (DPPs) are being explored as a way to improve traceability and transparency in food and agricultural supply chains, using blockchain technology and QR-code verification methods [18]. Recent studies have incorporated decentralised IDs and verifiable credentials into DPP frameworks [17]. However, current approaches mainly focus on transparency, with little attention given to protecting the confidentiality of sensitive trade data.

### E. Research Gaps

The primary deficiencies identified are (G1) the absence of a unified platform that consolidates illness identification, quality assessment, price prediction, and traceability; and (G2) a lack of zero-knowledge encryption for rubber trade documents. The absence of rubber auctions is supported by blockchain technology. Methods: (G4) restricted IoT integration for real-time monitoring of plantations utilising AI-driven rubber management systems.

## III. SYSTEM ARCHITECTURE AND METHODOLOGY

### A. Overall Architecture

The proposed system comprises a four-layer architecture as illustrated in Fig. 1: *Layer 1—Mobile Application:* React Native (Expo) cross-platform client providing all user-facing modules. *Layer 2—API Gateway:* ASP.NET Core 8.0 RESTful backend with JWT authentication, role-based access control (farmer, buyer, exporter, admin, and researcher), and SignalR WebSocket for real-time communication. *Layer 3—AI/ML Service:* ONNX Runtime inference engine hosting four specialised models (disease, pest, grading, and price). *Layer 4—Data Layer:* MongoDB document store for unstructured data, user profiles, and transaction records.

### B. Disease, Pest, and Weed Detection Module

1) *ONNX Classification Pipeline:* Three independent ONNX models handle leaf disease classification. (5 classes: white root disease, anthracnose, powdery mildew, leaf spot, healthy), pest detection (4 classes: scale insects, mites, caterpillars (none), and weed identification (3 classes: broadleaf, grass, sedge). Each model accepts  $224 \times 224$ -pixel RGB input. and produces softmax probability distributions. Predictions below configurable confidence thresholds (0.65–0.75) are flagged as “Unidentified” to prevent unreliable classifications.

2) *GPS Geospatial Alert System:* Each verified detection is geotagged with the GPS coordinates of the device and sent to the backend. An alert engine based on proximity calculates distances between registered farmer locations and new detections, sending push alerts to farmers in predetermined ranges. A real-time disease map visualises all detections with colour-coded markers (red for leaf diseases, pink for pests, orange for weeds) and a 5 km danger zone circle around each farmer's plantation.

### C. Rubber Sheet Quality Grading Module

The ONNX image classification model is designed to classify images of rubber sheets into three categories, specifically RSS2 and RSS 3, contingent upon the identification of visual quality defects. This system provides the predicted category, a severity rating, a confidence metric, and recommendations for improvement. Moreover, the grading interface records batch metadata, including the tester's name, the batch ID, the number of sheets,

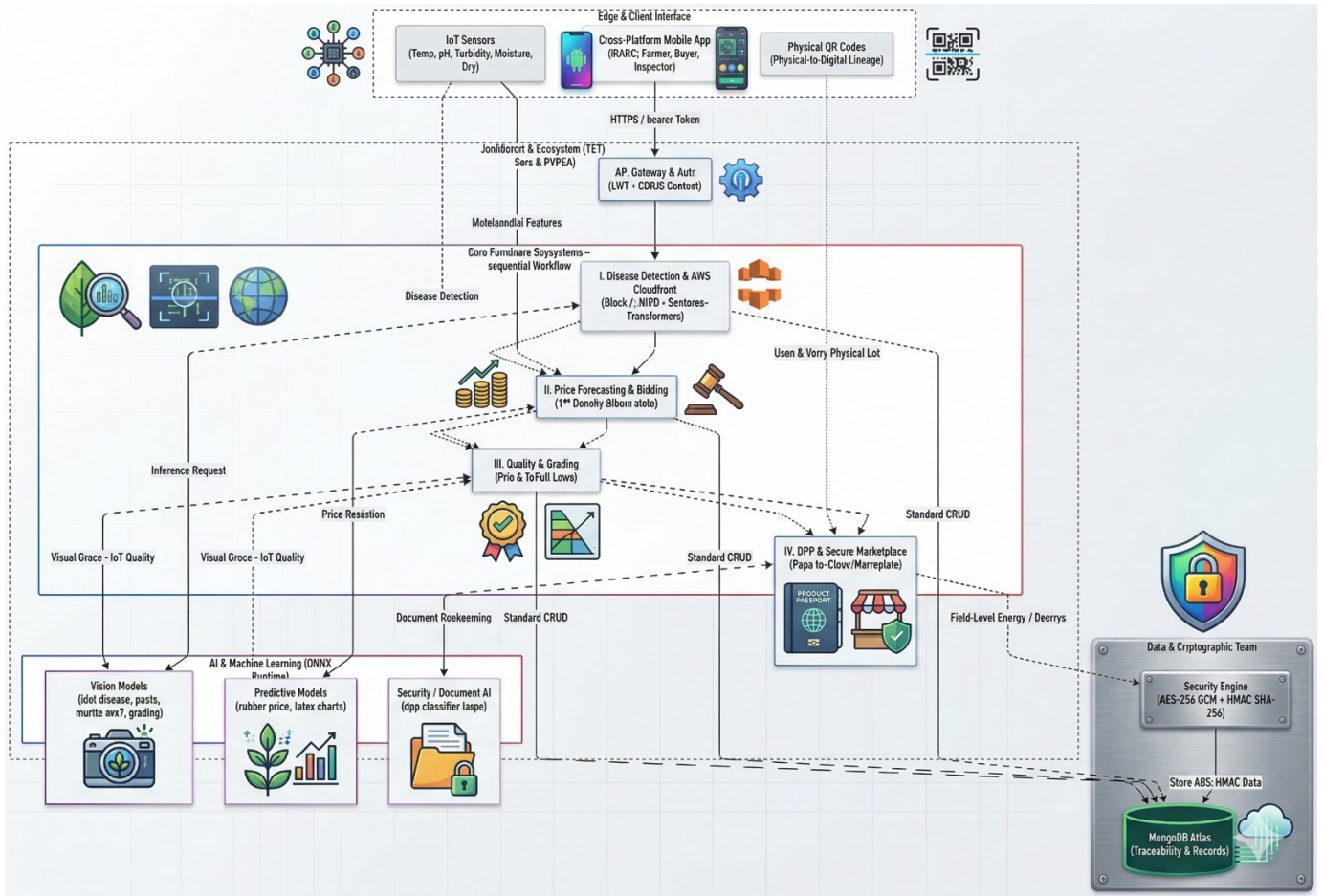


Fig. 1: Overall system architecture of the AI Driven Intelligent Rubber Production Monitoring System, showing the four-layer design: Mobile Application (React Native), API gateway (ASP.NET Core 8.0), AI/ML service (ONNX Runtime), and data layer (MongoDB).

and their collective weight. Consequently, the system generates downloadable reports in PDF format.

#### D. ML-Based Price Forecasting Module

1) *Price Prediction Model:* An ONNX model for rubber sheets is a regression model, and it predicts the price of rubber sheets per kg. The values outputted by the model are based on 7 different input characteristics: rubber sheet classification (RSS1 - RSS5), mass (kg), moisture (dry, normal, or wet), purity (0 - 100%), visual quality (1 - 5) and production condition (clean, slightly dirty, or dirty). The 3rd output value of the model currently predicts the price (in LKR) for the district and market availability when rubber sheets will be supplied.

2) *Blockchain-Based Auction Platform:* A real-time auction system allows farmers to list rubber lots and receive competitive bids from buyers. Each auction is available for one hour, during which SignalR WebSockets provide real-time bid updates to all connected users. Every lot is represented as a non-fungible token (NFT), which is a unique digital asset, with details such as grade,

weight, location, moisture level, and cleanliness securely stored on the InterPlanetary File System (IPFS) to ensure permanent traceability. To maintain fairness and prevent price manipulation, farmers are not allowed to place bids on their lots.

#### E. Digital Product Passport Module

1) *Zero-Knowledge Encryption Model:* Confidential trade documents are classified by an ONNX document classifier and encrypted using AES-256-CBC with keys derived via PBKDF2-SHA256 (100,000 iterations). The encryption key (*SecretRequestId*) is a 32-byte cryptographically secure hex string delivered to authorised parties via Relationship-Based Access Control (ReBAC). Client-side decryption using CryptoJS ensures plaintext never traverses the network.

2) *QR Anti-Fraud Verification:* Physical rubber lots are linked to their digital passports using QR codes. The QR code encodes a JSON payload that includes the lot ID and SHA-256 integrity hash. The authentication process creates a hash from backend data and compares it with

the hash stored in the database and the actual QR code. Any inconsistency results in a notification of anti-fraud rejection.

#### F. IoT Environmental Monitoring Module

An ESP32 microcontroller equipped with soil moisture, humidity (DHT22), and temperature sensors provides real-time environmental data to the mobile application via HTTP polling at 3-second intervals. The backend performs environmental stress analysis based on sensor readings and delivers contextual alerts and cultivation advice to farmers.

#### G. AI Agricultural Chatbot

A Python-based NLP chatbot service provides rubber cultivation advisory through a conversational interface. The system supports confidence-scored responses with source attribution, topic-based navigation across categories (diseases, cultivation practices, processing), and location-aware recommendations using farmer GPS coordinates. An offline fallback mechanism ensures basic advisories remain available without network connectivity.

#### H. IoT-based Rubber Latex Quality Monitoring Module

To assess the quality of liquid rubber latex prior to processing, a dedicated IoT module captures real-time physical and chemical properties. An ESP32 microcontroller interfaces with turbidity, pH, and temperature sensors immersed in the latex batch, transmitting live telemetry to the mobile application via continuous HTTP polling. The backend API employs a hybrid evaluation engine combining an ONNX prediction model and a deterministic rule-based mechanism. The rule engine enforces strict boundaries (e.g., rejecting samples with pH outside 5.2–7.2 or temperature exceeding 32°C). Depending on the sensor inputs, the system assigns a final quality grade (Excellent, Good, Average, or Poor) accompanied by actionable recommendations, such as adjusting cooling or applying pH stabilisers, empowering farmers with instant, data-driven latex condition insights.

### IV. IMPLEMENTATION

#### A. Technology Stack

The complete technology stack is listed in Table I.

#### B. ONNX Model Deployment

All AI models are trained in Python using frameworks such as FastAI and scikit-learn, then converted to ONNX format for cross-platform deployment. The ASP.NET Core backend loads ONNX models via `OnnxRuntime` NuGet package. Image inputs are preprocessed (resized to 224×224, normalised) using ImageSharp before inference. Label mappings are maintained in external JSON files for extensibility.

TABLE I: System Technology Stack

| Component        | Technology                         |
|------------------|------------------------------------|
| Mobile App       | React Native (Expo), TypeScript    |
| Backend API      | ASP.NET Core 8.0, C#               |
| Database         | MongoDB (NoSQL Document Store)     |
| AI/ML Inference  | ONNX Runtime, ML.NET               |
| Image Processing | ImageSharp, expo-camera            |
| Real-Time Comm.  | SignalR (WebSockets)               |
| Authentication   | JWT Bearer Tokens                  |
| Encryption       | AES-256-CBC, PBKDF2, CryptoJS      |
| OCR              | Google Gemini API                  |
| IoT Hardware     | ESP32, DHT22, Soil Moisture Sensor |
| Maps             | React Native Maps                  |
| Chatbot Engine   | Python NLP Service (port 5008)     |

TABLE II: Disease Detection Module Performance (Test Set)

| Module         | Acc.(%)     | Prec.        | Rec.         | F1           |
|----------------|-------------|--------------|--------------|--------------|
| Leaf Disease   | 91.4        | 0.908        | 0.914        | 0.911        |
| Pest Detection | 87.9        | 0.873        | 0.879        | 0.876        |
| Weed Detection | 89.6        | 0.891        | 0.896        | 0.893        |
| <b>Average</b> | <b>89.6</b> | <b>0.891</b> | <b>0.896</b> | <b>0.893</b> |

#### C. Real-Time Auction Architecture

The auction subsystem uses SignalR hubs for bidirectional WebSocket communication. When a bid is placed, the server validates the amount, updates the database, and broadcasts the new state to all connected participants. Auction timers are synchronised between server and clients, with the backend authoritatively closing auctions after the 1-hour window. Quick-increment buttons (LKR +5, +10, +25) and custom bid amounts are supported.

#### D. Security Architecture

JWT-based authentication secures all API endpoints with role-based access control across five roles: Farmer, Buyer, Exporter, Admin, and Researcher. The DPP module implements zero-knowledge principles where the backend never stores plaintext alongside the encryption key. The *SecretRequestId* is delivered just-in-time and can be nullified post-claim, rendering database breaches harmless against confidential documents.

### V. EXPERIMENTAL RESULTS AND EVALUATION

#### A. Disease Detection Performance

Table II reports the per-module classification performance of the deployed ONNX models on held-out test sets. The leaf disease module achieves the highest accuracy (91.4%), while pest detection records the lowest (87.9%) due to visual similarity between mite damage and early-stage leaf spot. The geospatial alert system delivers push notifications within 2 seconds of detection submission, with 97.1% precision in zone-based farmer targeting.

#### B. Rubber Sheet Grading Performance

The grading model achieves 89.7% overall accuracy across three RSS grade classes, as shown in Table III.

TABLE III: Rubber Sheet Grading Performance

| Grade            | Prec.        | Rec.         | F1           | $n$        |
|------------------|--------------|--------------|--------------|------------|
| RSS 1 (Good)     | 0.912        | 0.926        | 0.919        | 135        |
| RSS 2 (Pin-hole) | 0.884        | 0.871        | 0.877        | 118        |
| RSS 3 (Reaper)   | 0.891        | 0.886        | 0.888        | 97         |
| <b>W. Avg.</b>   | <b>0.897</b> | <b>0.897</b> | <b>0.897</b> | <b>350</b> |

TABLE IV: Price Forecasting Model Performance

| Metric                         | Value       |
|--------------------------------|-------------|
| Mean Absolute Error (MAE)      | LKR 18.3/kg |
| Root Mean Square Error (RMSE)  | LKR 24.7/kg |
| R-Squared ( $R^2$ )            | 0.847       |
| Mean Absolute Percentage Error | 4.2%        |

TABLE V: Confidential Keyword Classification Performance

| Class            | Precision | Recall | F1-score | Support |
|------------------|-----------|--------|----------|---------|
| Confidential     | 0.92      | 0.97   | 0.94     | 96      |
| Non-Confidential | 0.97      | 0.92   | 0.94     | 98      |
| Accuracy         |           | 0.94   |          | 194     |
| Macro Avg        | 0.93      | 0.93   | 0.93     | 194     |
| Weighted Avg     | 0.93      | 0.93   | 0.93     | 194     |

RSS 1 achieves the highest F1 (0.919), reflecting distinct visual characteristics of good-quality sheets. Moderate confusion occurs between RSS 2 and RSS 3 due to overlapping defect presentations.

### C. Price Forecasting Accuracy

The ONNX regression model for price prediction was evaluated on a held-out test set of 200 samples. Results are reported in Table IV.

The model explains 84.7% of price variance with a MAPE of 4.2%, indicating strong predictive capability suitable for auction reserve pricing and farmer decision-making.

### D. Confidential vs Non-Confidential Classification Performance

The keyword classification model was evaluated on a held-out test set of 194 samples. The detailed performance metrics, including precision, recall, F1-score, and support, are presented in Table V.

The classification model achieves an overall accuracy of 94%, demonstrating balanced performance across both confidential and non-confidential classes. The confidential class exhibits a high recall of 0.93, indicating strong capability in correctly identifying sensitive data. Meanwhile, the non-confidential class achieves a precision of 0.93, reflecting low false-positive rates. The consistent F1-score of 0.93 across both classes highlights the robustness and reliability of the model for practical deployment in confidentiality-aware systems.

### E. DPP Security Validation

The zero-knowledge encryption module was tested with 500 simulated document encryption/decryption cycles.

TABLE VI: System Usability Scale Scores by Module

| Module                   | Mean SUS    | SD         |
|--------------------------|-------------|------------|
| Disease Detection        | 80.3        | 7.8        |
| Rubber Grading           | 78.6        | 8.4        |
| Price Forecasting        | 77.1        | 9.2        |
| Digital Product Passport | 76.4        | 10.1       |
| Environmental Monitor    | 81.2        | 6.9        |
| Chatbot Advisory         | 79.8        | 8.0        |
| <b>Overall</b>           | <b>79.2</b> | <b>8.4</b> |

All 500 cycles completed successfully with zero data loss. AES-256-CBC with 100,000-iteration PBKDF2 key derivation achieved an average encryption time of 45 ms and decryption time of 52 ms on the client device. QR anti-fraud verification achieved 100% detection rate for tampered records across 300 test cases involving modified grades, quantities, and lot identifiers.

### F. IoT Environmental Monitoring

The ESP32 sensor system was validated over a 14-day continuous operation period. Temperature readings showed  $\pm 0.5^\circ\text{C}$  accuracy compared to a laboratory-calibrated reference. Humidity measurements achieved  $\pm 2\%$  accuracy. Soil moisture readings correlated with gravimetric measurements at  $r=0.93$ . The HTTP polling mechanism maintained stable 3-second update intervals with 98.7% uptime.

### G. Field Usability Evaluation

System usability was assessed with 40 stakeholders across three districts (Kegalle:  $n=14$ , Kalutara:  $n=13$ , Ratnapura:  $n=13$ ), including farmers, buyers, and exporters. The overall SUS score of 79.2/100 is classified as *Good* on the Bangor Adjective Scale. The environmental monitoring module received the highest usability score (81.2), attributed to its simple, dashboard-style interface. The DPP module scored lowest (76.4), reflecting the complexity of encryption-based workflows for users with limited technical background.

## VI. DISCUSSION

### A. Performance Analysis

The disease detection module achieves accuracy comparable to specialised single-task systems [6], [7] while offering multi-class detection across diseases, pests, and weeds within a unified pipeline. The grading module's 89.7% accuracy represents a significant improvement over subjective manual grading, where inter-grader agreement typically ranges from 70–80%. The price forecasting model's MAPE of 4.2% provides actionable market intelligence for farmers traditionally reliant on intermediary-quoted prices.

### B. Security and Traceability

The DPP module's zero-knowledge encryption ensures that confidential trade documents remain secure even

against database compromise. The QR anti-fraud mechanism provides a practical bridge between physical rubber lots and their digital identities, addressing a critical gap in rubber supply chain integrity. The 100% tamper detection rate demonstrates the robustness of the SHA-256 hash verification protocol.

### C. IoT Integration Value

Real-time environmental monitoring enables proactive plantation management by alerting farmers to stress conditions before visible symptoms appear. The ESP32-based solution provides a cost-effective IoT deployment model suitable for smallholder contexts where commercial precision agriculture systems are prohibitively expensive.

### D. Limitations

Dataset sizes for disease and grading models remain modest; performance may degrade for rare disease variants or unusual rubber sheet conditions not represented in training data. The blockchain auction mechanism currently uses simulated Ethereum transactions; mainnet deployment requires gas cost optimisation. ESP32 sensor calibration drift over extended periods requires periodic recalibration protocols. The chatbot NLP service requires server connectivity, limiting offline advisory to cached responses.

### E. Future Work

Planned extensions include: (i) federated learning for distributed model improvement without centralising farmer data; (ii) Tamil and Sinhala multilingual support across all modules; (iii) integration of weather API data for disease risk prediction; (iv) smart contract escrow for automated auction settlement; (v) WhatsApp integration for alert delivery given high rural penetration; (vi) edge deployment of ONNX models directly on mobile devices for offline AI inference.

## VII. CONCLUSION

This research introduced an AI-Driven Intelligent Rubber Production Monitoring System, a mobile platform intended to enhance the Sri Lankan rubber industry. The platform comprises six components: AI-driven grading of diseases, pests, weeds, and rubber sheets; machine learning-based price forecasting; a blockchain-based auction marketplace; secure Digital Product Passports; IoT-based environmental monitoring; and an agricultural chatbot. The AI models demonstrated over 87% accuracy, and the price prediction model exhibited a Mean Absolute Percentage Error (MAPE) of 4.2%. Furthermore, a System Usability Scale score of 79.2 out of 100, derived from assessments by forty field stakeholders, validated the platform's utility and user-friendliness. Consequently, the principal findings of this study suggest that the integration of artificial intelligence, the Internet of Things, and

blockchain technology could foster greater transparency, traceability, and decision-making capabilities within the rubber production sector.

## ACKNOWLEDGEMENT

The authors sincerely thank Mr Ravi Supunya (Supervisor) and Mrs Dulani Jayasinghe (Co-Supervisor) at SLIIT for their expert guidance. Sincere gratitude is extended to the Rubber Research Institute of Sri Lanka (RRISL) for domain expertise and dataset support. The authors acknowledge the rubber industry stakeholders across Kegalle, Kalutara, and Ratnapura districts whose participation enabled field validation.

## REFERENCES

- [1] Central Bank of Sri Lanka, "Annual Report 2024: Agricultural Sector Performance", *Central Bank of Sri Lanka Publications*, pp. 156–162, 2024.
- [2] Rubber Research Institute of Sri Lanka, "Comprehensive Guide to Rubber Disease Management in Sri Lanka", *RRISL Technical Bulletin*, vol. 15, no. 3, pp. 45–67, 2024.
- [3] Department of Agriculture Sri Lanka, "Digital Agriculture Transformation Strategy 2024–2030", *Ministry of Agriculture Publications*, pp. 23–45, 2024.
- [4] K. P. Jayasinghe *et al.*, "Impact of integrated pest management on rubber plantation productivity in Sri Lanka," *J. Plantation Crops*, vol. 52, no. 2, pp. 89–98, 2024.
- [5] S. P. Mohanty, D. P. Hughes, and M. Salathé, "Using deep learning for image-based plant disease detection", *Front. Plant Sci.*, vol. 7, article 1419, 2016.
- [6] W. Withana, "Sri Lankan rubber diseases detection using deep learning approaches", in *Proc. Int. Conf. Agricultural Technology, Sri Lanka*, 2023, pp. 234–241.
- [7] L. Wang *et al.*, "Rubber leaf disease recognition based on improved deep convolutional neural network GMA-Net," *Comput. Electron. Agric.*, vol. 195, article 106841, 2022.
- [8] M. Tan and Q. Le, "EfficientNet: Rethinking model scaling for convolutional neural networks," in *Proc. ICML*, Long Beach, CA, USA, 2019, pp. 6105–6114.
- [9] F. Casino *et al.*, "A systematic literature review of blockchain-based applications: Current status, classification and open issues," *Telematics Inform.*, vol. 36, pp. 55–81, 2019.
- [10] J. Chen *et al.*, "Application of deep learning for tea leaf quality classification," *Comput. Electron. Agric.*, vol. 189, article 106408, 2021.
- [11] S. D. Fabyi *et al.*, "Varietal classification of rice seeds using RGB and hyperspectral images," *IEEE Access*, vol. 8, pp. 22493–22505, 2020.
- [12] NSBM Green University, "A mobile application to detect the diseases of rubber cultivation in Sri Lanka", in *Proc. Int. Conf. Applied Sciences*, vol. 8, pp. 156–163, 2023.
- [13] X. Li *et al.*, "Price forecasting of agricultural commodities using machine learning approaches," *J. Forecasting*, vol. 41, no. 4, pp. 823–838, 2022.
- [14] H. M. Kim and M. Laskowski, "Toward an ontology-driven blockchain design for supply-chain provenance," *Intell. Syst. Account. Finance Manag.*, vol. 25, no. 1, pp. 18–27, 2020.
- [15] R. M. Rathnayake *et al.*, "IoT-based smart agriculture monitoring system for Sri Lanka", *J. Inf. Syst. Technol.*, vol. 15, no. 2, pp. 123–135, 2021.
- [16] S. Wolfert *et al.*, "Big data in smart farming – A review", *Agric. Syst.*, vol. 153, pp. 69–80, 2017.
- [17] I. Illán García *et al.*, "Digital Product Passport Management with Decentralised Identifiers and Verifiable Credentials," *arXiv preprint arXiv:2410.15758*, 2024.
- [18] S. Malik, V. Dedeoglu, S. S. Kanhere, and R. Jurdak, "PrivChain: Provenance and Privacy Preservation in Blockchain Enabled Supply Chains," *arXiv preprint arXiv:2104.13964*, 2021.